The Microsystems and Microelectronics (MEMS) faculty at IIT Bombay are pursuing an exceptionally wide and deeply specific array of micro- and nano-scale device, system, and integration technologies. Collectively, their current and recent research projects include: development of ultra-high-Q silicon-photonic microresonators and integrated electro-optomechanical modulators for on-chip coherent communications; graphene- and 2D-material-based nanomechanical resonant-mass sensors for single-cell mechanophenotyping and disease biomarker detection; CMOS-compatible piezoelectric AlN thin-film energy harvesters and vibratory microgenerators optimized via topology-aware finite-element design for self-powered IoT nodes; microcantilever arrays with plasmonic and biofunctional coatings for multiplexed point-of-care cancer diagnostics; microfluidic organ-on-chip platforms integrating microvalve and micropump networks for liver- and lung-tissue–vessel co-culture and drug-toxicity screening; wafer-level, low-temperature anodic-bonded and glass-frit hermetic packaging techniques for harsh-environment pressure and inertial sensors operating up to 600 °C; microscale gas-chromatography columns fabricated in silicon and glass with embedded micro-heaters and integrated chemoresistive sensor arrays for real-time environmental monitoring; surface-micromachined electrostatic RF-MEMS switches and tunable capacitors designed for insertion loss < 0.3 dB and power-handling > 2 W in 5G front-ends; flex-circuit–bonded, stretchable microsensor skin arrays employing silicone-encapsulated piezoresistive pressure and temperature sensors for human–machine interaction and soft-robotics feedback; SiGe-based three-axis MEMS gyroscopes with noise floors below $0.01°/\sqrt{h}$ achieved via mechanical tuning-fork designs and vacuum-packaged SOI fabrication; micro-thruster arrays for CubeSat attitude control using colloidal-propellant vaporization in microchannel reactors; UV-lithography–defined SU-8 microactuators for tactile haptic interfaces and micro-gripper assemblies; CMOS-MEMS integration of capacitive accelerometers with on-chip SAR-ADC and dynamic-range compression algorithms for automotive crash detection; and additive-manufactured micro-lattice structures for acoustic metamaterials and high-efficiency microthermal transport in microreactors. This superset captures the full depth and breadth of MEMS research at IIT Bombay, spanning novel materials, device architectures, system integration, and application-specific packaging and reliability strategies.